

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE) /

Seventh Semester of B. Tech. Examination (CE)

April-2018

CE316 Artificial Intelligence (A.I.) / CE407 Artificial Intelligence (A.I.)

Date: 05.04.2018, Thursday Time: 01:30 p.m. To 04:30 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

- Q-1 Answer the following questions.**
- a) What is the Turing test? What is the significance of it? **03**
 - b) Differentiate: Supervised and Unsupervised learning **04**
 - c) What are the advantages and disadvantages of Hill Climbing Search Technique? **04**
- Q-2 Answer the following questions. (Any Three)** **12**
- [A] Write a PROLOG program to find last element in the list.
 - [B] What are the characteristics of the problem that are to be analyzed when choosing an appropriate method to solve the problem? Explain.
 - [C] Transfer following into FOPL (First Order Predicate Logic):
 - 1. All employees earning more than 50,000 Rs. /year, pay income tax.
 - 2. No employee of University earns more than the VC.
 - [D] Write a PROLOG program for performing mathematical operations like addition, multiplication, division and subtraction of two numbers.
- Q-3 Answer the following questions. (Any Two).** **12**
- [A] What are the various steps involved in NLP.
 - [B] Write down the water jug problem. Give the rules for water jug problem.
 - [C] What is the main drawback of back-propagation network? What is the reason of it? What do you mean by learning rate? Explain how it is affecting learning process?

SECTION-II

- Q-4 Answer the following questions.**
- a) Differentiate between Forward Reasoning and Backward Reasoning **03**
 - b) Compare Procedural Vs Declarative knowledge. **04**
 - c) What are the uses of fuzzy logic in AI? What are the criticisms for fuzzy logic? **04**
- Q-5 Answer the following questions. (Any Three)** **12**
- [A] Write a note on: Dempster Shafer Theory
 - [B] Draw the semantic network for the following sentence:
'Every dog in the town has bitten the constable.'
 - [C] Explain the principles of means_end_analysis approach to problem solving.
 - [D] Consider the following set of sentences:
 - 1. Marcus was a man
 - 2. Marcus was a Pompeian

3. Marcus was born in 40 A.D.
4. All men are mortal
5. All pompeians died when the volcano erupted in 79 A.D.
6. No mortal lives longer than 150 years.
7. It is now 2005

Answer the following question with the help of proof by resolution:

Is Marcus alive?

Q-6 Answer the following question. (Any Two)

12

- [A] Trace the execution of the constraint satisfaction procedure in solving the crypt-arithmetic problem:

SEND+MORE=MONEY

- [B] Draw the block diagram of a typical Expert System. Which are the types of users involved with Expert System? Also compare it with traditional software system.
- [C] Why do game playing programs work from a current state to a goal rather than backward from a goal? What properties of a game might suggest using a backward strategy?